CSE7850/CX4803 Machine Learning in Computational Biology



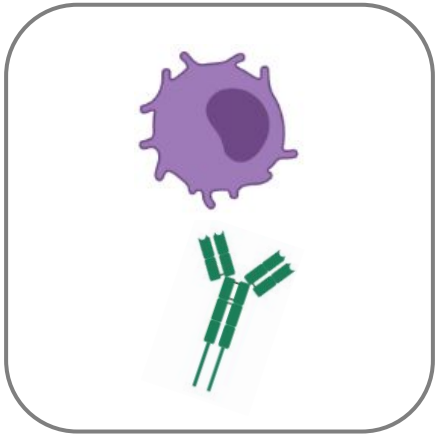
ML4BIO Hackathon:

ML-guided protein engineering with active learning

Yunan Luo

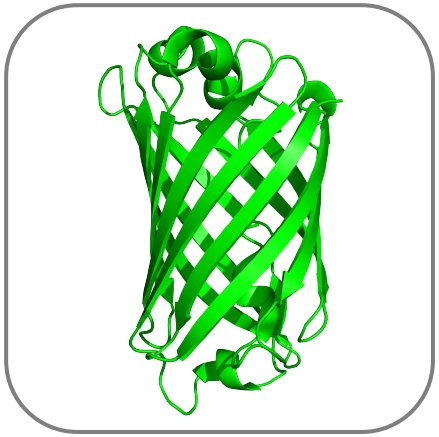
Protein engineering

Applications in biology and medicine



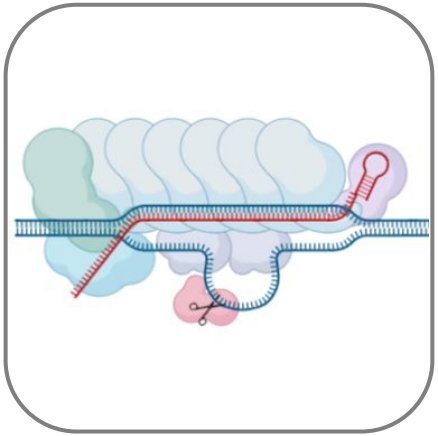
Antibody

Binding affinity ⬆



Fluorescent protein

Fluorescence ⬆

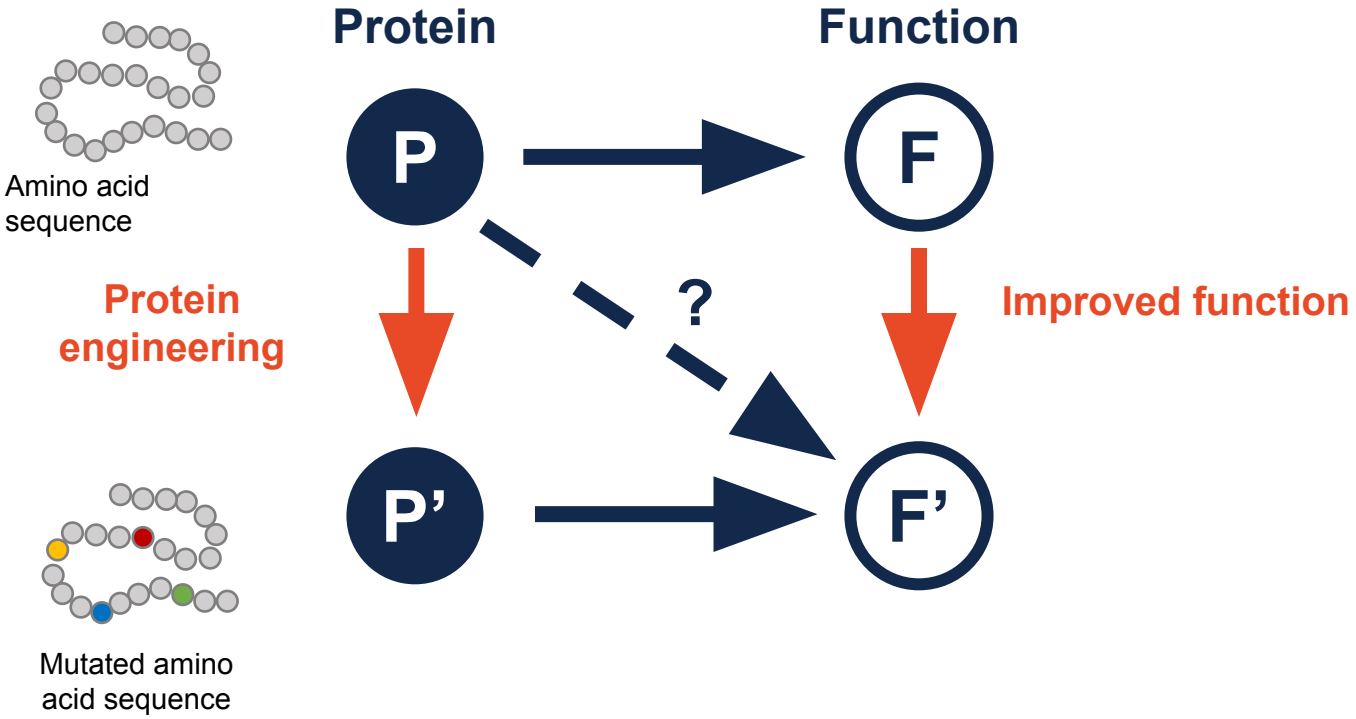


CRISPR/Cas9

Specificity ⬆

Protein engineering

Optimizing natural proteins

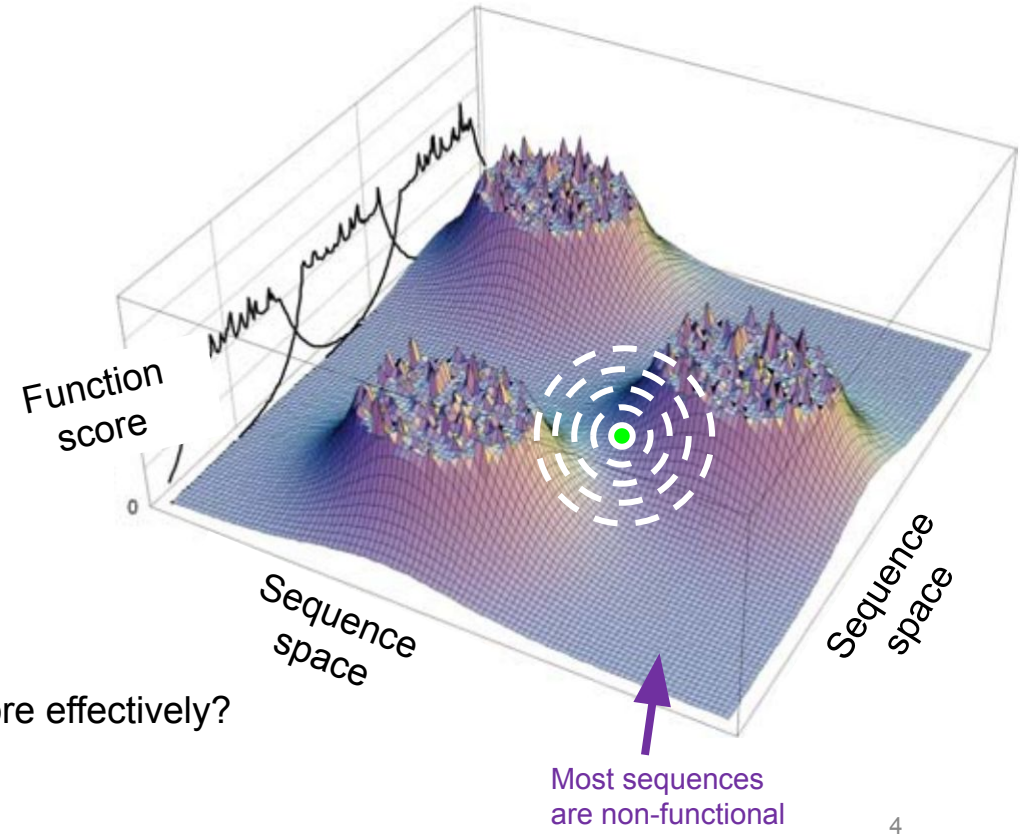


Images adapted from Ahmed Badran, MIA Lecture, 2017

Search proteins in the sequence space

Huge sequence space

5 amino acid protein: ~10^{6} sequences
10 amino acid protein: ~10^{13} sequences
50 amino acid protein: ~10^{65} sequences
100 amino acid protein: ~10^{130} sequences
300 amino acid protein: ~10^{325} sequences



How to search through the sequence space more effectively?

Image from Hayashi et al., 2006

Protein fitness landscape

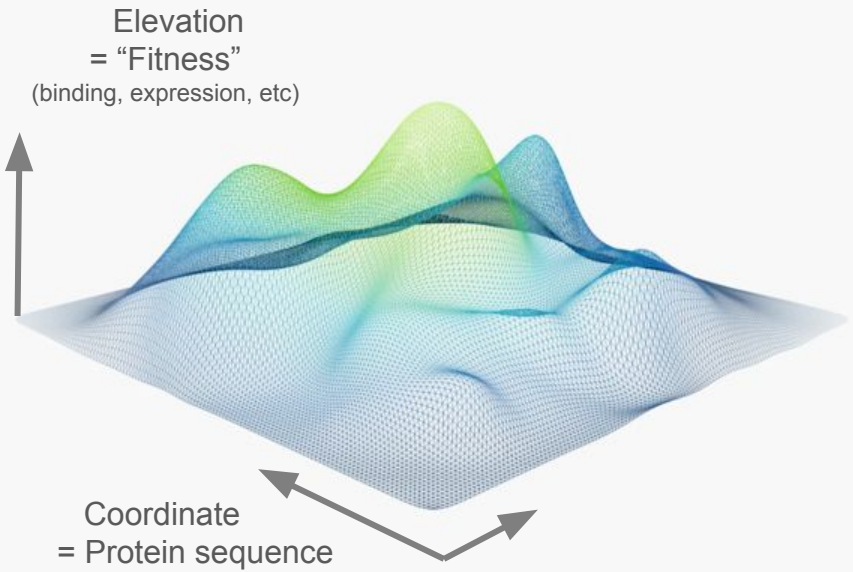
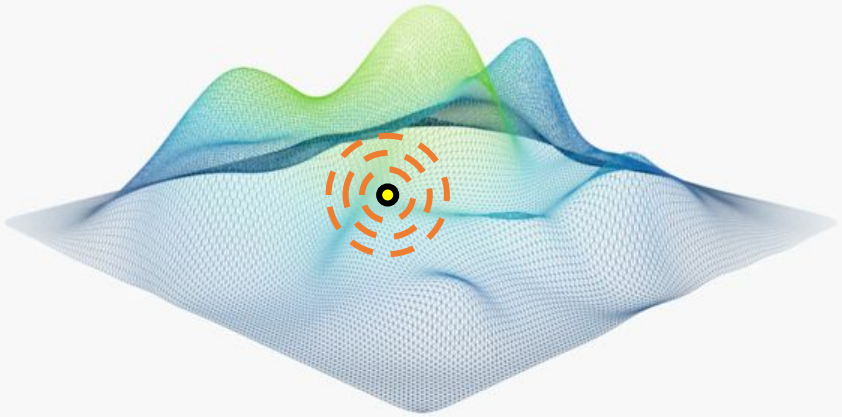


Figure: cradle.bio

How to find functional proteins?

Protein fitness landscape

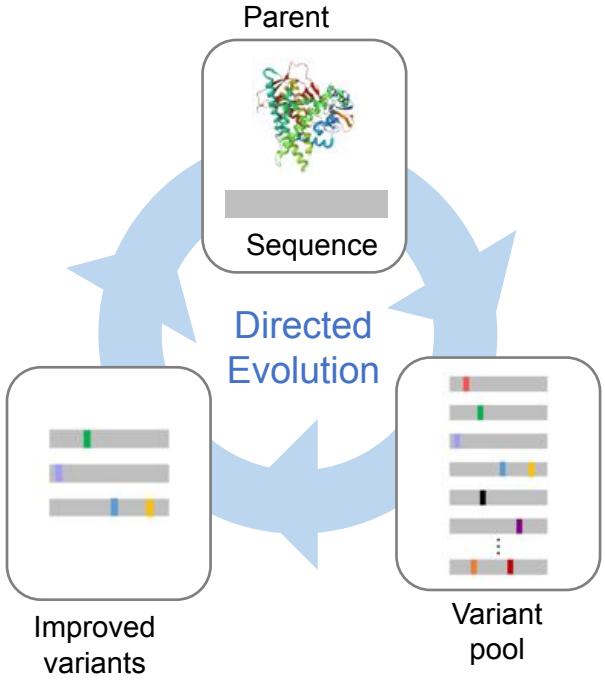


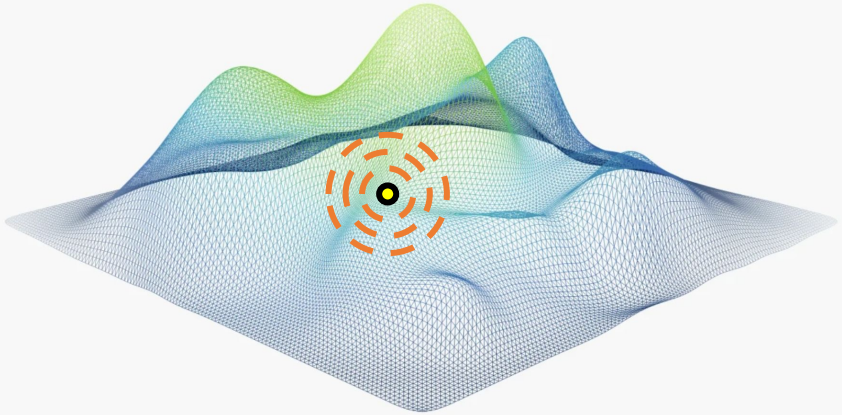
How to find functional proteins?

Directed evolution for protein engineering
(2018 Nobel Prize in Chemistry)



Frances Arnold
(Caltech)





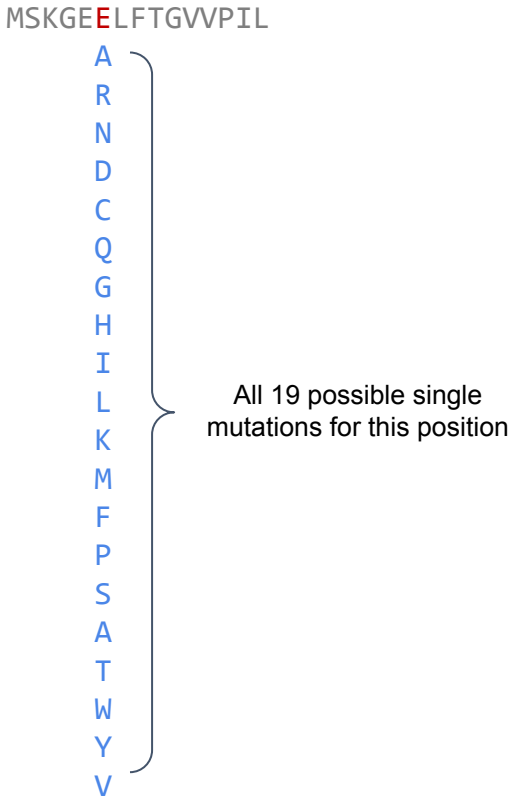
Protein engineering

- Mutagenesis experiments

Sequence	Fitness of interest (e.g., binding affinity, stability, expression)
MSKGEELFTGVVPIL	1.0
MSKCEELFTGVVPIL	0.5
MSKGWELFTGVVPIL	5.4
MSKGEELFTSVVPIL	0.6
MSKGEELFTVVVPIL	1.1
MSYGEEVFTGVVPIL	2.5
MSKGEFGFAGVVPIL	0.1
MDYGEELFTVVVPSL	1.3
MSKGWELFTGVAPIL	6.1

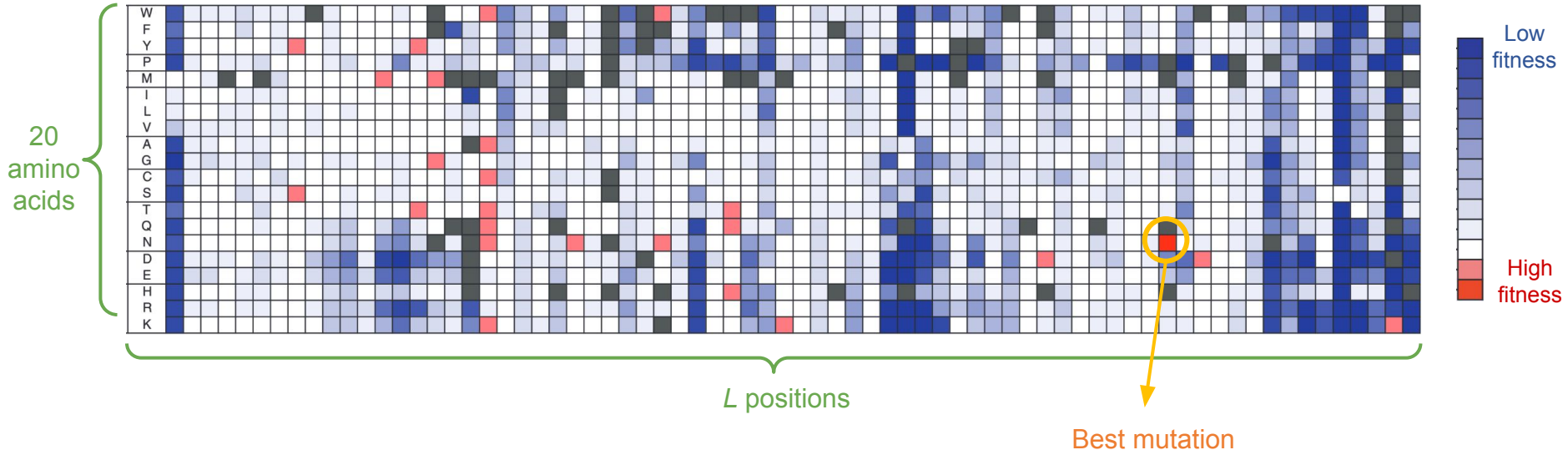
Function characterization requires significant experimental effort

Let’s consider single mutation

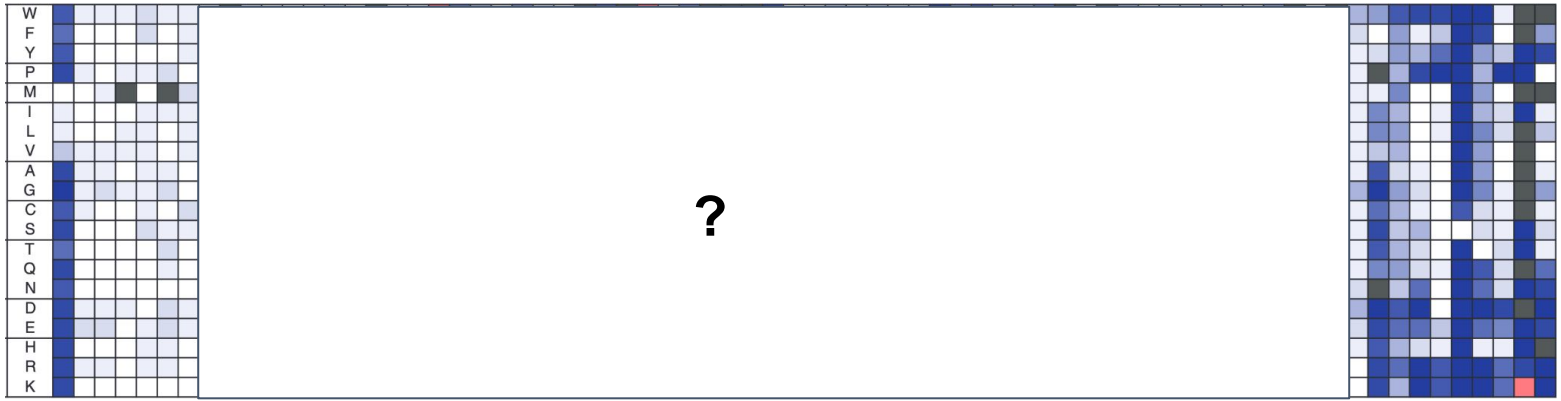


Deep mutational scanning (DMS)

- In wet-lab experiments, we can measure the fitness all possible single mutations for a protein with length *L*

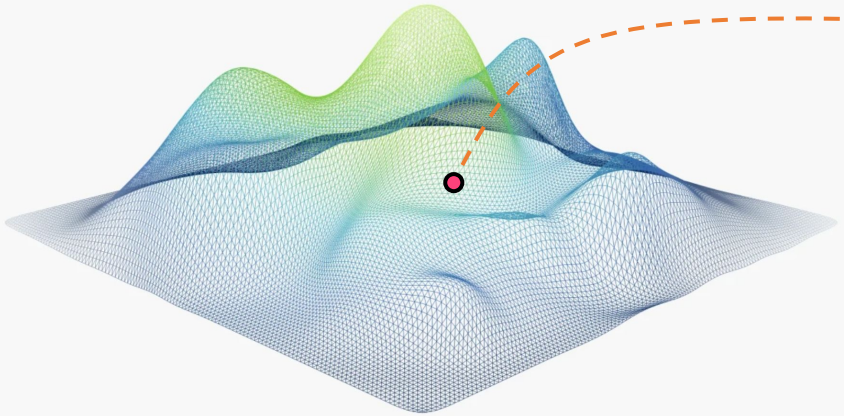


What if the DMS data is incomplete?

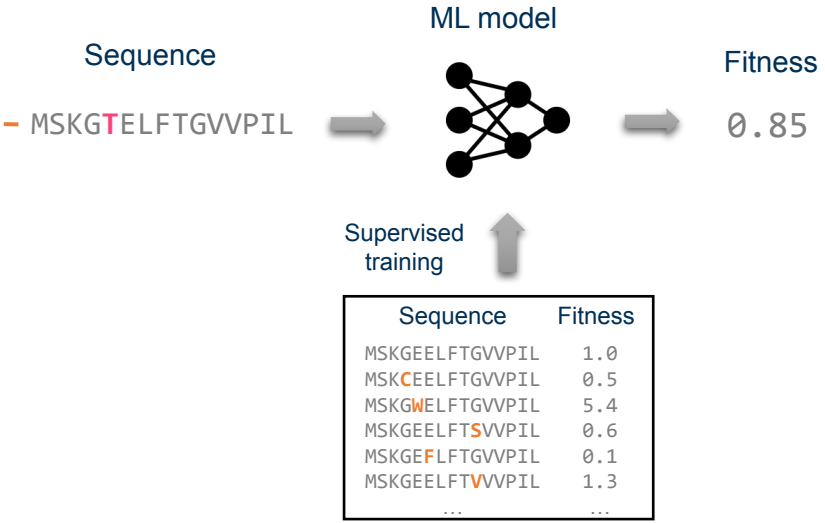


How would you find the best mutation?

ML-guided protein engineering

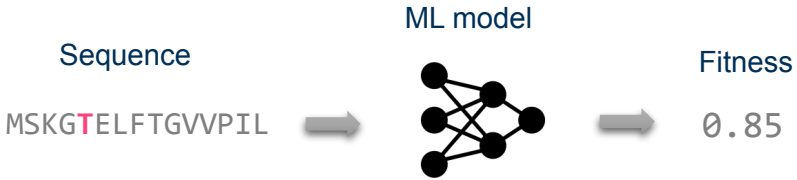


- **Sequence-to-fitness prediction** with machine learning



Task 1: Develop a ML model to predict protein fitness from sequence

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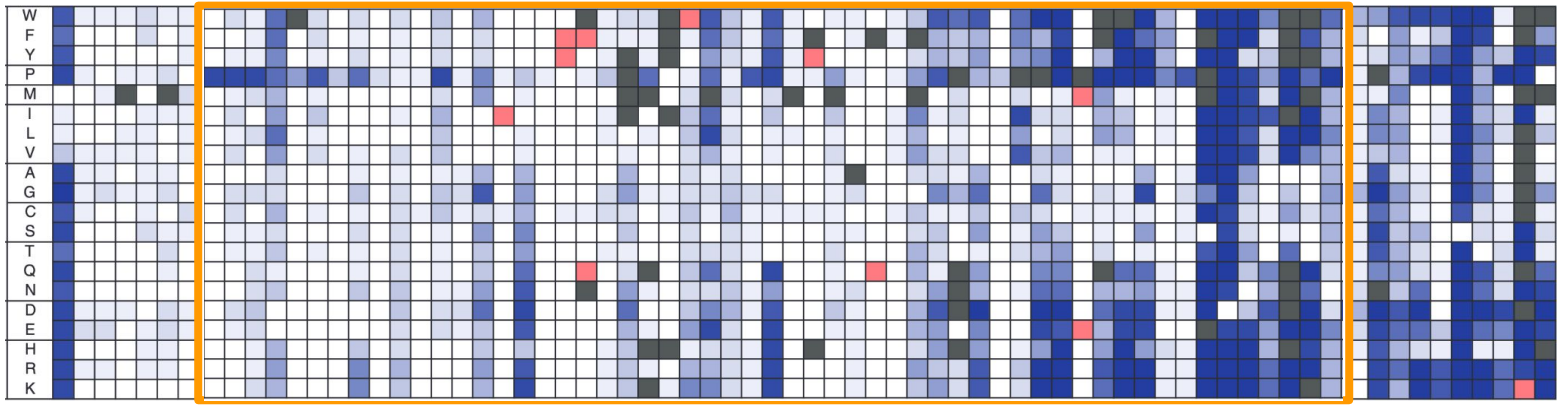


Sequence encoding:
- One-hot?
- Protein language model embedding?
- …

Model architecture:
- Regression model?
- Random forest?
- Fully-connected neural network?
- CNN? RNN? Transformer?

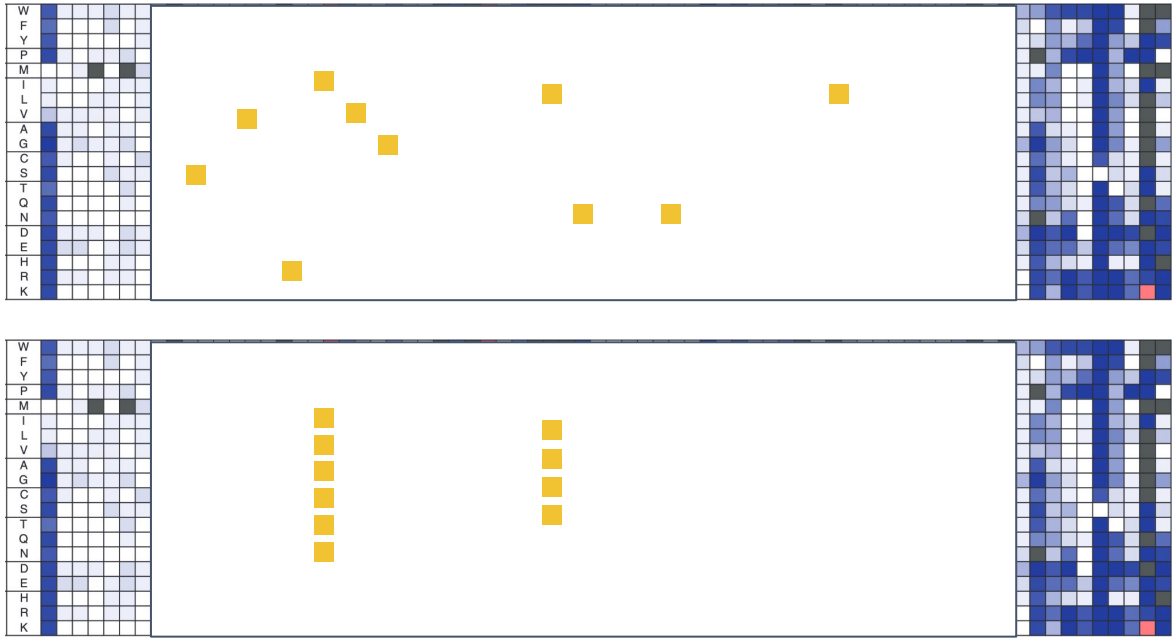
Other strategies:
- Hyperparameter tuning?
- Finetuning pre-trained model?

Model prediction



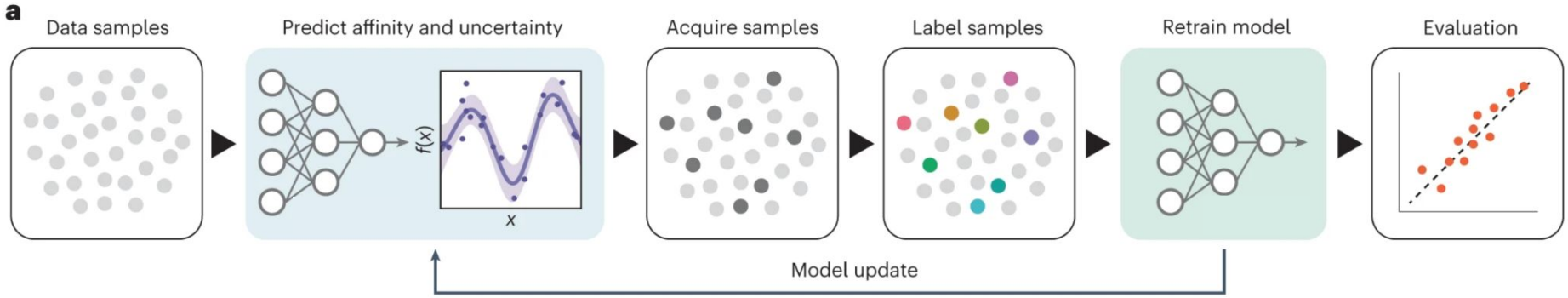
How to improve the ML model?

- Use more training data!
- Suppose you can run a wet-lab experiment to measure the fitness for 10 more mutants, which 10 would you query that can most improve your ML model?

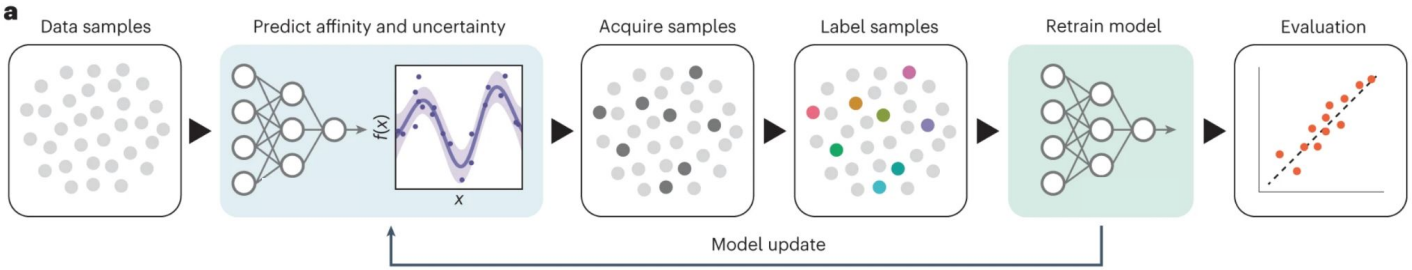


Improve your ML model through active learning

Active learning: a learning algorithm can interactively query a human user (or some other information source), to label new data points with the desired outputs, and update/retrain the model using newly labeled data



How to select data points to label?

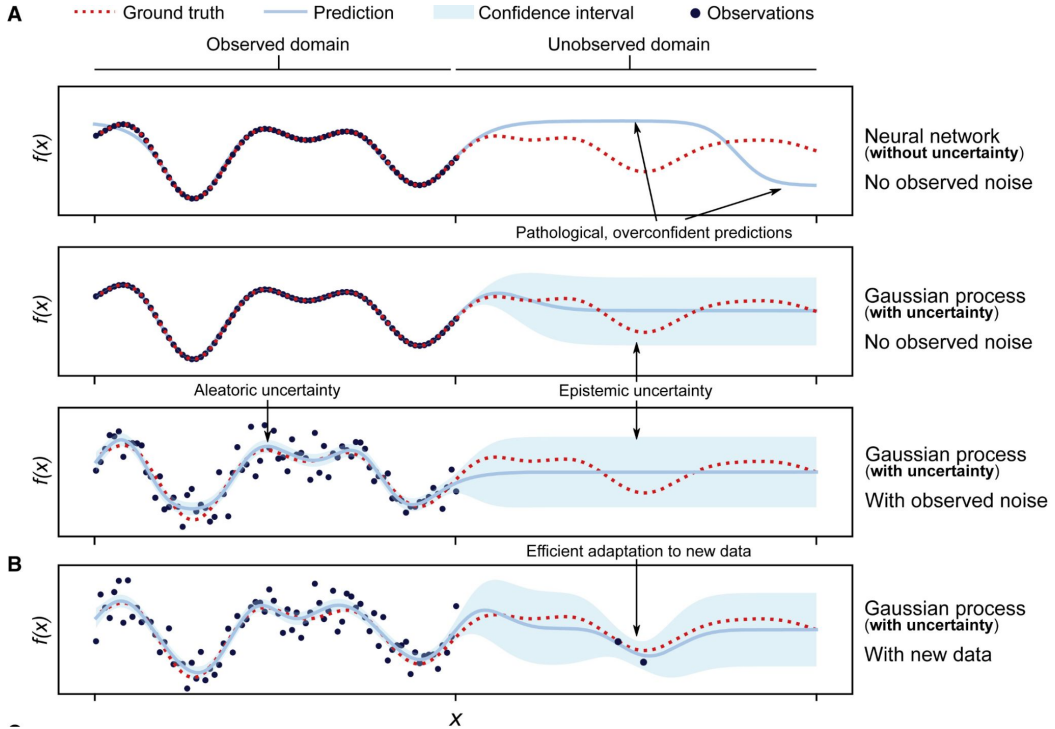


Intuition: select the data points such that, once labeled, can most improve ML model performance

- Random?
- Greedy? (e.g., select the top-10 predicted mutations)
- Others?

Example: Uncertainty quantification

- Instead of acquiring top-predicted data points to label, select those that can most address ML model's uncertainty

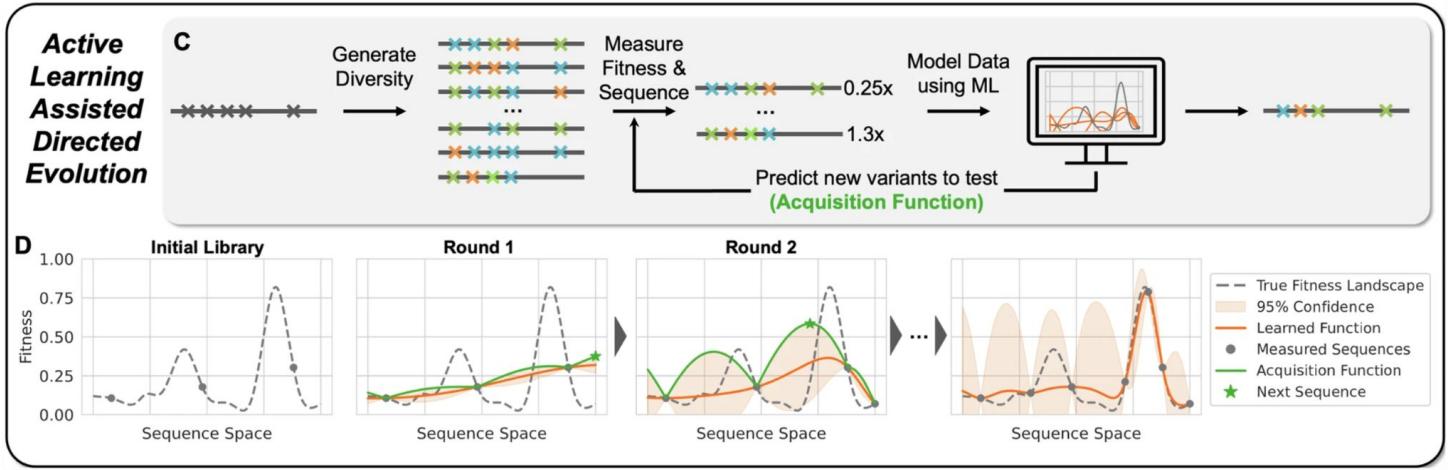


https://doi.org/10.1016/j.cels.2020.09.007

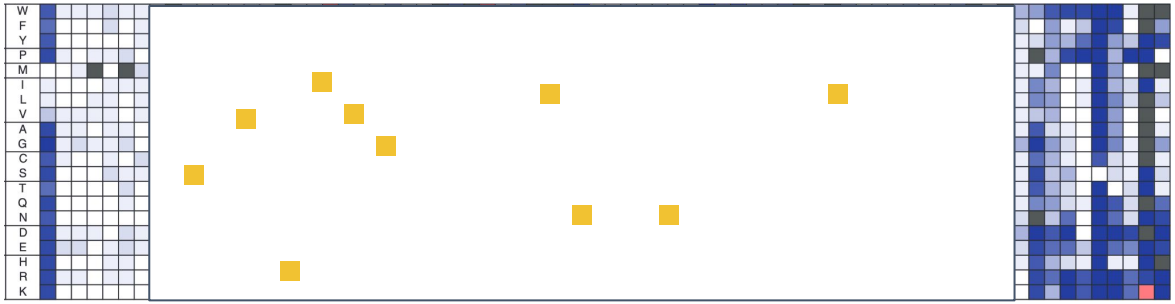
Acquisition function

- How to select data points to label is an active research problem. In ML literature, researchers have defined a set of “acquisition functions” that score/rank data points to be labeled

- Paper #1 [Active learning-assisted directed evolution | Nature Communications]
  - Closely related to this Hackathon
  - Evaluated a set of classical acquisition functions (they may or may not work well for this Hackathon)



Task 2: active learning to improve your fitness prediction model



- We provide an query oracle (online)
- You can send mutants to this oracle (up to 100 per query), and we return you their fitness value
- You can make up to 3 queries
- How to wisely make the query to most improve your ML model?

Hackathon

- Tasks:
  - Develop an ML model to predict protein fitness from sequence
  - Design a good strategy to query more data and refine your ML model to improve the prediction performance
- Evaluation:
  - 5%: Early checkpoint submission
  - 45%: Overall prediction accuracy (correlation)
  - 15%: Effectiveness of identifying the best mutations in your top-10 prediction list
  - 35%: Report (2-4 pages)
- More details in the Hackathon document

Top Percentage Recalled	Points Awarded
1%	3 (Full Credit)
2%	2.5
5%	2
10%	1.5
20%	1
50%	0.5
100%	0 (No Credit)